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Research paper



# Evaluating choice-based demand management strategies for day-to-day home delivery planning

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ABSTRACT

As demand for online shopping and home delivery increases rapidly, courier companies often offer customer-oriented services, placing severe constraints on delivery planning and making routes inefficient. This study presents a framework for evaluating strategies to balance customer satisfaction and delivery efficiency in day-to-day home delivery planning. We first model the online shopping frequency choices by an ordered logit model and delivery timing choices by a nested logit model and estimate them using stated choice data. Based on the day-to-day delivery demand simulated by these models, we optimize a quasi-dynamic multi-period vehicle routing problem with a variable routing period ($d$-MPVRP) and evaluate the delivery efficiency under various demand management strategies. Our results show that different demand management strategies, such as surcharges for highly-demanded time slots and discounts for frequent users, have different effectiveness across urban and rural contexts. The discount on the flexible delivery option for customers who frequently shop online allows consolidation of multiple delivery orders, and was particularly effective in rural areas. Our results suggest that effective demand management in day-to-day home delivery requires strategies considering geographic and demand characteristics to achieve delivery efficiency while maintaining customer satisfaction.

## 1. Introduction

The rapid growth of e-commerce (EC) worldwide has drastically increased home delivery demand, impacting last-mile delivery operations. The efficiency of last-mile delivery routes is affected by spatiotemporal demand distribution patterns. Not only can demand concentration on specific days or time slots intensify the delivery workload, but small demands spatially and temporally separated from other demands can also lead to inefficient delivery due to low carry rates or detours. With the proliferation of EC, customers may easily order small quantities of goods more frequently, spurring delivery inefficiency. Therefore, to achieve efficient and sustainable last-mile delivery, it is desirable to disperse demand while at the same time consolidating it to a moderate degree.

Demand management for last-mile delivery has been intensively studied in the literature of attended home delivery (AHD) (Waßmuth et al., 2023). The AHD studies mainly consider pricing and allocation of delivery time slots for demand management by influencing the choice behavior of customers (Abdollahi et al., 2023; Campbell & Savelsbergh, 2006; Mackert, 2019; Strauss et al., 2021; Yang & Strauss, 2017; Yang et al., 2016). While these studies focus on scheduled delivery, real-world EC marketplaces often offer customers other delivery options that are more flexible for couriers. For example, Amazon's normal (standard) delivery allows a courier to decide on which day and what time of day within a certain period to deliver a good to a customer (Oyama et al., 2024). That is, a courier may postpone a delivery request considering efficiency, for example, when the customer is located separately from the other customers scheduled on a day. This flexible delivery option also allows for the consolidation of multiple delivery orders of the same customer if their delivery periods overlap. These strategies are only possible through day-to-day home delivery planning.

However, day-to-day home delivery has not been sufficiently studied in the literature. Although some researchers study a multi-period vehicle routing problem (MPVRP) with due dates, which can consider flexible delivery spanning multiple days like normal delivery (Archetti et al., 2015; Estrada-Moreno et al., 2019; Larrain et al., 2019), there is no existing study that evaluates day-to-day demand management strategies by integrating choice models with an MPVRP framework.

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The objective of this study is to evaluate demand management strategies for day-to-day AHD, explicitly considering customers' choice behavior. In the day-to-day context, the same customers may request multiple deliveries. Therefore, we focus on (1) online shopping frequency choice behavior and (2) delivery timing choice behavior, including the choices of delivery options, dates, and time slots. Using discrete choice models, we simulate the changes in these behavior in reaction to various demand management strategies, including a longer delivery time window and the pricing of certain delivery options/slots. In addition, considering the online shopping frequency, we design a strategy that incentivizes customers with high frequency to choose normal delivery, allowing for the consolidation of multiple deliveries of the same customers. These strategies are evaluated in terms of both delivery efficiency and customer satisfaction. To quantify delivery efficiency, we present a quasi-dynamic MPVRP that considers new orders appearing and updates multi-period route planning every day. The day-to-day delivery efficiency considers total routing cost, vehicle operation cost, and holding cost that is required to postpone the delivery. Using this simulation-based framework, we evaluate and compare different choice-based demand management strategies. The evaluation is conducted in two different spatial demand settings, representing urban and rural areas, to investigate the effects of the differences in online shopping frequency and demand density on the performance of the strategies.

The contribution of this paper lies in the development of a framework for evaluating choice-based demand management strategies for day-to-day AHD, considering different delivery options, namely next-day, scheduled, and normal deliveries. Since previous studies on AHD focus on time slot management (Agatz et al., 2011; Klein et al., 2017; Mackert, 2019; Yang et al., 2016), our day-to-day AHD framework that considers different delivery options and order frequency presents an important contribution to the literature. The evaluation framework is divided into demand simulation and route optimization. Demand simulation integrates two choice models of online shopping frequency and delivery timing, allowing us to predict the change in choice behavior in reaction to various day-to-day demand management strategies that are difficult to evaluate in existing AHD models. We estimate these choice models using stated choice data with a representative sample of EC customers. While some previous studies perform detailed analysis of order frequency and delivery timing choice behavior by estimating advanced models (Dias et al., 2020; Oyama et al., 2024; Schaefer & Figliozzi, 2021), they do not evaluate the impact of choice behavior on delivery operations. In contrast, we integrate choice models with a routing optimization model to evaluate the impact of behavioral changes on delivery. As for routing optimization, we propose a flexible day-to-day routing model based on a quasi-dynamic MPVRP that considers daily appearing demand. The effectiveness of this routing policy is validated through a comparison with the original MPVRP (Archetti et al., 2015) that assumes an ideal condition where all the demand for the entire planning period is known. Through numerical experiments, we show the effectiveness of our framework in evaluating various choice-based demand management strategies for day-to-day AHD. Our results offer valuable insights for decision-makers and marketplaces, highlighting the importance of planning delivery for multiple days and considering the behavior of customers.

The remainder of the paper is organized as follows. Section 2 reviews the literature relevant to this study. Section 3 describes our framework, including the data, choice models, and a quasi-dynamic MPVRP model. Section 4 present model estimation results and numerical results. We then discuss the results in Section 5 and finally conclude our study in Section 6.

## 2. Literature review

In this section, we review the studies relevant to this study, focusing on the topics of (1) choice-based delivery demand management, (2) delivery timing choice models, and (3) MPVRP models.

### 2.1. Choice-based delivery demand management

The AHD problem integrates demand management with a vehicle routing problem (Fleckenstein et al., 2023; Waßmuth et al., 2023). Campbell and Savelsbergh (2005) is the first paper to examine both aspects of demand management and delivery operations. They investigated a dynamic routing and scheduling problem to determine whether to accept an order and in which time slot a vendor will make a delivery. In contrast, choice-based demand management explicitly models customers' choice behavior affected by delivery attributes, e.g., the price or availability of time slots. Campbell and Savelsbergh (2006) investigated incentives for customers to choose time windows with a lower insertion cost, which is the increase in the routing cost when the customer is added to the delivery route. They showed that the discount of delivery in such time slots would lead to a reduction in delivery costs. As such, time slot pricing has been examined in various studies and has shown its effectiveness in increasing revenues (Akkerman et al., 2022; Asdemir et al., 2009; Klein et al., 2017; Koch & Klein, 2020; Lebedev et al., 2021; Ulmer, 2020; Yang & Strauss, 2017; Yang et al., 2016). Strauss et al. (2021) showed that combining flexible time windows in time slot pricing can effectively reduce delivery costs. In addition, Abdollahi et al. (2023) recently attempted to offer the best time slot by incorporating order forecasting. As a result, their work achieved a considerable profit improvement over (Yang et al., 2016) and Yang and Strauss (2017). However, they also showed a decrease in the number of slots offered to customers as the delivery date approached, which reduced the desirability of the customer choice situation. Other demand management methods include longer delivery time windows, which may reduce customers' convenience but can significantly reduce delivery costs (Köhler et al., 2020; Strauss et al., 2021). Furthermore, customizing the time slots offered to different customers can also be effective in securing revenue (Lang et al., 2021; Mackert, 2019). Although these demand management approaches can contribute to improving profits, it is still not clear if high profits can be achieved without reducing customer satisfaction.

In addition, most AHD models focus on time slot management for scheduled delivery, and flexible delivery timing options that span multiple days, like normal delivery, have not yet been studied. Although (Abdollahi et al., 2023) presented demand management in the delivery date choice context, the literature on demand management for day-to-day home delivery is still limited.

### 2.2. Delivery timing choice behavior models

Predicting the change in delivery timing choice of EC customers is important for demand management of home delivery, and time slot choice behavior is often studied. Campbell and Savelsbergh (2006) used a simple linear time slot choice model to capture the impact of pricing on choice probability. Asdemir et al. (2009) extended this work by using a multinomial logit (MNL) model to analyze the reaction of customers' time slot choice to delivery service attributes. Yang et al. (2016) and Yang and Strauss (2017) estimated an MNL model using real e-grocer data and showed the effectiveness of time slot pricing in revenue management. Strauss et al. (2021) constructed a nested logit (NL) model with nests defined by flexible and standard slots. However, these choice models often included only customers' sensitivity to price and the dummy effects of specific slots. In contrast, Agatz et al. (2021) showed that labeling some time slots as "green" increased the propensity of environmentally friendly customers to choose such time slots. Amorim et al. (2024) modeled the choice of time slots that span multiple days and include different slot sizes. By doing so, the authors considered the attributes of delivery time and slot size, in addition to price.

While time slot choice models for scheduled delivery have been extensively presented in the literature, those models fail to consider a flexible delivery option for couriers. The only exception is Oyama

et al. (2024), who integratedly modeled the choice of delivery options, dates, and time slots. The delivery options considered include normal delivery, which allows couriers to select the delivery timing within a multiple-day period, as well as next-day delivery and scheduled delivery. They revealed that customers may change their choices to normal delivery when other delivery options are expensive or inconvenient in terms of slot size. By estimating a mixed logit model with a semi-nonparametric approach, Oyama et al. (2024) also flexibly analyzed heterogeneous preferences among customers. However, they did not evaluate the impact of behavioral changes on the efficiency of delivery operations.

In addition, although the order frequency of EC customers has been extensively studied (e.g., Dias et al., 2020; Pani et al., 2024; Reiffer et al., 2023; Saphores & Xu, 2021), a frequency model has not been explicitly integrated in day-to-day home delivery demand management. In the context of day-to-day home delivery, the same customers may request multiple deliveries within a short period, which impacts delivery planning.

This study estimates both a delivery timing choice model of Oyama et al. (2024) and an order frequency choice model using data from the same sample. While we use ordered logit and nested logit models that do not incorporate unobserved heterogeneity, their tractability allows us to embed them within a day-to-day routing optimization framework and evaluate the operational impact of customer behavioral changes.

### 2.3. Multi-period vehicle routing problem

Studies on MPVRP can be categorized into static and dynamic approaches. Regarding the static problem, Archetti et al. (2015) introduced an MPVRP with delivery orders, which do not specify delivery date but only have due dates, and showed that extending the due date can reduce delivery costs. Their objective function includes a holding cost to postpone deliveries, which balances the trade-off with the delivery distance. However, static MPVRPs (Archetti et al., 2015; Estrada-Moreno et al., 2019; Larrain et al., 2019) assume that the demand for the entire planning period is fully known in advance, making them difficult to apply to real-world last-mile delivery operations. To address this limitation, several studies presented dynamic MPVRPs (e.g., Keskin et al., 2023; Subramanyam et al., 2021; Wen et al., 2010), considering dynamic order arrivals over time. These dynamic MPVRPs often employ a rolling horizon framework, where previously determined routes may be updated when new orders arrive (e.g., Cordeau et al., 2015; Markov et al., 2020).

Despite these advancements, no existing study has integrated customers' choice behavior of delivery option or time slot into MPVRPs. Our study explicitly models customer choices of delivery options and time slots, spanning multiple days, and integrates them into an MPVRP model. We also present a quasi-dynamic MPVRP that is more computationally efficient than dynamic problems and allows for integration into a simulation-based framework. By doing so, our study presents a more realistic and effective demand management approach that balances customer satisfaction and operational efficiency.

## 3. Materials and methods

### 3.1. Stated choice data

To analyze and model the choice behavior of EC users, we utilized data from a survey conducted between April and May 2021. The respondent is a member of a large Japanese courier service company. The respondents were among "Kuroneko Members", members of Yamato Holdings Co., Ltd. which has the largest share of parcel delivery in Japan. This survey included responses from 4872 individuals (response rate of 4.87%), and we obtained both revealed preference (RP) and stated preference (SP) data. For details on the survey design and data, see Oyama et al. (2024).

**Table 1**
Frequency of online shopping among respondents ($N = 4721$).

| Ordinal variable | Frequency | Number of respondents | Percentage (%) |
|---|---|---|---|
| 1 | Once a month or less | 1803 | 38.2 |
| 2 | 2–3 times a month | 858 | 18.2 |
| 3 | Once or twice a week | 1997 | 42.3 |
| 4 | 3–4 times a week | 63 | 1.3 |

**Table 2**
Delivery option choice results.

| | | Number of choices | Percentage (%) |
|---|---|---|---|
| Delivery option ($N = 19,860$) | Next-day delivery | 8429 | 42.4 |
| | Scheduled delivery | 7166 | 36.1 |
| | Normal delivery | 4265 | 21.5 |
| Delivery slot ($N = 7166$) | Morning | 3262 | 45.5 |
| | Night | 1822 | 25.4 |
| | Other | 2082 | 29.1 |
| Delivery date ($N = 7166$) | Weekday | 1327 | 18.5 |
| | Holiday | 5839 | 81.5 |

We cleaned the response data of 151 respondents who had no experience with online shopping and classified the frequency of online shopping of valid respondents into the following four categories:

The SP survey was conducted on the assumption that respondents would purchase the same products again that they purchased online the previous time. Respondents were asked to select a delivery option among (1) next-day delivery, (2) scheduled delivery (2 to 8 days after order, from 9:00 a.m. to 9:00 p.m.), and (3) normal delivery (date and time cannot be specified). Respondents who selected scheduled delivery were additionally asked to select delivery date and time. Each respondent makes one choice for each of the five different choice scenarios. For each scenario, three attributes were varied: delivery fee, delivery time (the number of days required for delivery), and range of time slots (2, 3, or 4 h). For example, if the range of time slots is 2 h, there are 6 (12/2) choices of slots. In addition to the response data from those who have never used EC as mentioned above, we excluded from our analysis the data from the choosers of the dominated alternatives, resulting in a final sample size of 19,860.

Table 2 shows an aggregate result, the percentage of each delivery option being chosen. Approximately 42% of observations chose next-day delivery, and among those for scheduled delivery, 46% selected morning delivery. Note that the survey period coincided with "Golden Week" (a major holiday period in Japan), which resulted in approximately 82% of observations choosing a holiday to receive the ordered items.

### 3.2. Choice models

#### 3.2.1. Online shopping frequency model

We now formulate an online shopping frequency model using an ordered logit model. The ordered logit model introduces the latent variable $Y_n^*$ for individual $n$, defined as:

$$Y_n^* = \sum_k \gamma_k X_{nk} + \epsilon_n \tag{3.1}$$

where $\gamma_k$ represents the $k$th parameter to be estimated, $X_{nk}$ the individual-specific attribute, and $\epsilon$ an i.i.d. Gumbel-distributed error term. The frequency choice $Y_n$ is the ordered response in Table 1, and the relationship between $Y_n$ and $Y_n^*$ is expressed as follows:

$$\begin{cases} Y_n = 1 & \text{if } Y_n^* \le \kappa_1 \\ Y_n = 2 & \text{if } \kappa_1 \le Y_n^* \le \kappa_2 \\ Y_n = 3 & \text{if } \kappa_2 \le Y_n^* \le \kappa_3 \\ Y_n = 4 & \text{if } \kappa_3 \le Y_n^* \end{cases} \tag{3.2}$$

Here, $\kappa$ are the threshold parameters to be estimated. In the ordered logit model, the probability of choosing each interval of $Y_n^*$ is considered as the choice probability. With $Z_n$ as the linear combination of

the explanatory variables for an individual $n$, these choice probabilities can be formulated as follows:

$$P(Y_n = 1) = \frac{1}{1 + \exp(Z_n - \kappa_1)} \tag{3.3}$$

$$P(Y_n = 2) = \frac{1}{1 + \exp(Z_n - \kappa_2)} - \frac{1}{1 + \exp(Z - \kappa_1)} \tag{3.4}$$

$$P(Y_n = 3) = \frac{1}{1 + \exp(Z_n - \kappa_3)} - \frac{1}{1 + \exp(Z - \kappa_2)} \tag{3.5}$$

$$P(Y_n = 4) = 1 - \frac{1}{1 + \exp(Z_n - \kappa_3)} \tag{3.6}$$

The model is estimated by maximum likelihood estimation. The likelihood function is defined as

$$\mathcal{L}_F(\gamma, \kappa) = \sum_{n=1}^{N} \sum_{i \in \{1,2,3,4\}} \delta_{ni} \ln P(Y_n = i) \tag{3.7}$$

where $\delta_{ni}$ equals 1 when individual $n$ selects online shopping frequency $i$. By maximizing the above likelihood function $\mathcal{L}_F$, we simultaneously estimate the unknown parameters $\gamma$ and the threshold parameters $\kappa$.

#### 3.2.2. Delivery timing choice model

We next formulate a model for EC users' delivery timing choices. As mentioned earlier, there are three delivery options in the choice model: next-day, scheduled, and normal delivery. For scheduled delivery, the delivery time is classified into earliest (morning), latest (evening), and others, which are combined with the delivery date to form the alternatives. Thus, there are 21 ($7 \times 3$) time slot alternatives for scheduled delivery.

Based on the random utility maximization (RUM) theory (McFadden, 1986), individual $n$ is assumed to choose an alternative $i$ that maximizes the utility

$$U_{ni} = V_{ni} + \epsilon_{ni}, \tag{3.8}$$

where $V$ is the systematic utility. We apply the NL model with the alternatives of scheduled delivery as a lower nest. This is based on the assumption that the error terms in the utilities of the 21 alternatives for scheduled deliveries are correlated. In our model, let $G$ and $S$ be the set of alternatives belonging to the upper and lower nests, respectively. That is, denoting 2+ as 2 days after the order, we have

$$\begin{aligned} G &= \{\text{next day delivery}, \text{scheduled delivery}, \text{normal delivery}\}, \\ S &= \{(2+, \text{morning}), (2+, \text{night}), (2+, \text{other}), \ldots, (8+, \text{night})\}. \end{aligned} \tag{3.9}$$

The probability $P_n(g, i)$ that individual $n$ chooses delivery option $i \in S$ in nest $g \in G$ is defined as

$$\begin{aligned} P_n(g, i) &= P_n(i \mid g) P_n(g) \\ &= \frac{\exp(\mu_g V_{ni})}{\sum_{i' \in S} \exp(\mu_g V_{ni'})} \frac{\exp(\mu \Lambda_{ng})}{\sum_{g' \in G} \exp(\mu \Lambda_{ng'})} \end{aligned} \tag{3.10}$$

where $\mu_g$ is a scale parameter associated with the subgroup of scheduled delivery. We assume that the scale $\mu$ is standardized as 1. Therefore, only $\mu_g$ is estimated as the scale parameter. The logsum variable $\Lambda_{ng}$ is given by

$$\Lambda_{ng} = \frac{1}{\mu_g} \ln \sum_{i \in S} \exp(\mu_g V_{ni}). \tag{3.11}$$

The systematic utility $V_{ni}$ is defined as

$$V_{ni} = \beta_{\text{day}} \cdot \text{day}_i + \beta_{\text{fee}} \cdot \text{fee}_i + \delta_{\text{i, scheduled}} \cdot \beta_{\text{window}} \cdot \text{window}_i + \boldsymbol{\alpha}\boldsymbol{\sigma} \tag{3.12}$$

where $\text{day}_i$ is the number of days required to be delivered when choosing $i$, $\text{fee}_i$ is the delivery fee of $i$ in JPY, and $\text{window}_i$ is the time window of scheduled delivery, i.e., the length of a time slot in hours, which implies how long the individual has to be at home at maximum to receive the ordered item. The dummy variable $\delta_{i,\text{scheduled}}$ is 1 if $i$ is an option for scheduled delivery and 0 otherwise. The parameter $\boldsymbol{\alpha}$ is a vector of the alternative specific constants, and $\boldsymbol{\sigma}$ is 1 if alternative $i$ belongs to each delivery option, date and time. For example, the systematic utility of the alternative specifying scheduled delivery in the morning time slot two days after the order date (on a holiday) is

$$V_{n,2+,\text{morning}} = \cdots + \alpha_{\text{scheduled}} + \alpha_{\text{holiday}} + \alpha_{\text{morning}} \tag{3.13}$$

The log-likelihood function of the delivery timing choice model is given by

$$\mathcal{L}_D(\beta, \mu) = \sum_{n=1}^{N} \sum_{t=1}^{T_n} \left[\ln P_n(i_{nt} \mid g_{nt}) + \ln P_n(g_{nt})\right] \tag{3.14}$$

where $i_{nt}$ and $g_{nt}$ are the chosen alternative and the nest by individual $n$ at choice task $t$, respectively. We estimate the parameter $\beta$ and $\mu_g$ by maximizing the log-likelihood function $\mathcal{L}_D$.

### 3.3. Multi-period vehicle routing with day-to-day demand

#### 3.3.1. Quasi-dynamic MPVRP

In this subsection, we present a quasi-dynamic MPVRP that takes into account vehicle capacity, time windows, and due dates, to evaluate day-to-day delivery planning. We assume that the courier plans the routes of vehicles at the beginning of each day, given the delivery requests from customers that have been revealed by the end of the previous day. Since this study does not consider same-day delivery, delivery for orders that appear on the day will be scheduled in the delivery plan for the next day or later.

In reality, demand information is revealed on a day-to-day basis, and delivery routes are re-planned accordingly. Therefore, planning delivery routes over multiple days under uncertain future demand is not practically possible. On the other hand, optimizing delivery routes of a day considering only the demand to be delivered on the day may result in inefficient delivery when delivery to some customers can be postponed. For example, in Fig. 1(a), the blue customer is far from the other customers on day $t$, while s/he is close to some other customers on day $t+1$. Therefore, a route would become more efficient by postponing the delivery to the blue customer to day $t+1$ as in Fig. 1(b).

To address this issue, this study proposes to (re)plan delivery routes every day for $d$ days from that day. In this study, we call $d$ a *routing period*. When $d$ is 1, only the routes of the day are planned, which means that a single-period VRP is solved every day. When $d$ is greater than 1, the courier plans routes for multiple days from days $t$ to $t + d - 1$. Under this routing policy, the courier can choose not to deliver to some customers on a day and instead on or after the day as long as ensuring the due dates of the requests (e.g., when customers choose normal delivery whose delivery date is not specified). This courier's decision is influenced by holding costs, which are required to postpone delivery and store the parcels at a delivery office. Note that when $d$ is equal to the total number of days in the planning period and the demand information is initially and fully known, our quasi-dynamic MPVRP corresponds to the original MPVRP in Archetti et al. (2015).

#### 3.3.2. Formulation

Let $i, j \in \{0, 1, \ldots, n\}$ be indices representing nodes, where 0 denotes depot and 1 to $n$ denote customers. We assume that there is a single depot for each planning area and that all vehicles start at the depot. Then, let $c_{ij}$ be the travel cost (distance) between nodes $i$ and $j$. Also, let $u_{ik}$ be the amount of vehicle loaded when vehicle $k$ leaves node $i$, $t_{ik}$ be the arrival time of vehicle $k$ at node $i$, $m$ be the maximum number of vehicles that can be operated on each day, $Q$ be the capacity of a vehicle, and $v$ be the initial cost of the vehicle. For customer $i \in \{1, \ldots, n\}$, the holding cost of the package is $h_i$, the quantity of demand is $q_i$, the specified delivery time (i.e., time window) is within the range of $(e_i, l_i)$, and the range of possible delivery dates is from $s_i$ to $f_i$ ($s_i = f_i$ for next day delivery and scheduled delivery).

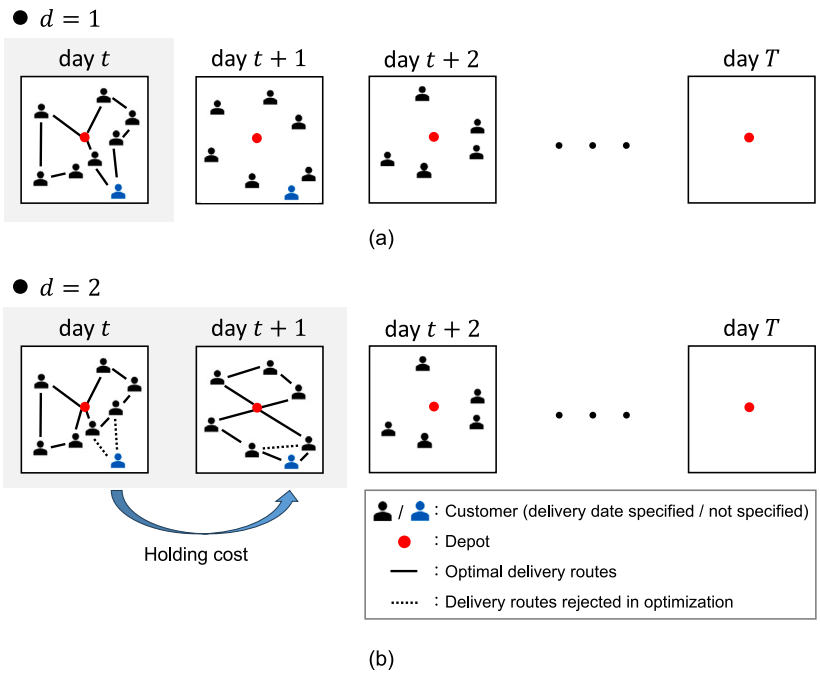



**Fig. 1.** Difference in route planning by $d$.

The decision variable of the problem is:

$$x_{ijk}^{s} = \begin{cases} 1 & \text{if vehicle } k \text{ travels from nodes } i \text{ to } j \text{ on day } s, \\ 0 & \text{otherwise,} \end{cases}$$

and the MPVRP with a variable routing period ($d$-MPVRP) solved on day $t$, i.e., the route planning of days $t$ to $t+d-1$ given the demand revealed by day $t$, is formulated as follows:

[$d$-**MPVRP**] (day $t$)

$$\min_{x} z(x) \equiv v \sum_{j=1}^{n} \sum_{s=t}^{t+d-1} \sum_{k=1}^{K} x_{0jk}^{s} + \sum_{i=0}^{n} \sum_{j=0}^{n} \sum_{s=t}^{t+d-1} \sum_{k=1}^{K} c_{ij} x_{ijk}^{s} + \sum_{i=1}^{n} \sum_{s=t+1}^{t+d} h_i \Big(1 - \sum_{s'=t}^{s-1} \sum_{j=0}^{n} \sum_{k=1}^{K} x_{ijk}^{s'}\Big), \tag{3.15}$$

subject to

$$\sum_{k=1}^{K} \sum_{j=0}^{n} \sum_{s=t}^{f_i} x_{ijk}^{s} = 1 \qquad \forall i \in \{1,2,\dots,n\}, \tag{3.16}$$

$$\sum_{j=0}^{n} x_{ijk}^{s} = \sum_{j=0}^{n} x_{jik}^{s} \quad \forall i \in \{0,1,2,\dots,n\}, \forall s \in \{1,2,\dots,T\}, \forall k \in \{1,2,\dots,K\}, \tag{3.17}$$

$$\sum_{j=1}^{n} \sum_{k=1}^{K} x_{0jk}^{s} \le m \quad \forall s \in \{1,2,\dots,T\}, \tag{3.18}$$

$$u_{ik} - u_{jk} + Q \cdot x_{ijk}^{s} \le Q - q_j \quad \forall s \in \{1,2,\dots,T\}, \forall k \in \{1,2,\dots,K\}, \forall i,j \in \{1,2,\dots,n\},\ i \ne j, \tag{3.19}$$

$$t_{ik} - t_{jk} + M \cdot x_{ijk}^{s} \le M - c_{ij} \quad \forall s \in \{1,2,\dots,T\}, \forall k \in \{1,2,\dots,K\}, \forall i,j \in \{1,2,\dots,n\},\ i \ne j, \tag{3.20}$$

$$q_i \le u_{ik} \le Q \quad \forall i \in \{0,1,2,\dots,n\}, \forall k \in \{1,2,\dots,K\}, \tag{3.21}$$

$$e_i \le t_{ik} \le l_i \quad \forall i \in \{0,1,2,\dots,n\}, \forall k \in \{1,2,\dots,K\}. \tag{3.22}$$

The objective function (3.15) of the problem [$d$-MPVRP] is the sum of vehicle costs, transportation costs, and inventory holding costs in the order they are presented in the formula. The holding cost is added for each day delivery is postponed.[1] Constraint (3.16) ensures that delivery is made by $f_i$, the due date of delivery to customer $i$. Constraint (3.17) ensures that the number of vehicles arriving at and departing from customer $i$ on each day is equal. Constraint (3.18) imposes a constraint that the number of vehicles departing from a depot on each day is at most $m$. Inequalities (3.19) and (3.20) are called the potential (MTZ) formulation (Miller et al., 1960) and ensure subtour elimination, where $M$ in inequality (3.20) represents a sufficiently large constant. Inequalities (3.21) and (3.22) impose a vehicle capacity constraint and a delivery time window constraint, respectively. Note that constraints (3.20) and (3.22) can be removed by adding a penalty term for early arrival and late delivery to the objective function, as in Archetti et al. (2015) and Keskin et al. (2023).

Under this routing policy, the problem [$d$-MPVRP] is solved every day $t$ given the demands revealed by the day, and the delivery routes for $d$ days from day $t$ are re-planned in a rolling horizon manner. Note that $d$ is a hyperparameter and does not change dynamically during the planning period. Therefore, we compare routing policies with different values of $d$ in the numerical experiments of Section 4.2.

### 3.4. Evaluation framework

We evaluate several demand management strategies through numerical simulations using the choice models and VRP that are defined in the previous subsections. The evaluation framework is summarized in Fig. 2.

As shown in Fig. 2, the planning period spans 24 days from Day 1 (Saturday) to Day 24 (Monday). To eliminate the influence of customers generated before the planning period, we set the evaluation period for a strategy to one week from Day 10 (Monday) to Day 16 (Sunday).

Demand for each simulation run is generated as follows. We consider a fixed number of customers whose locations are randomly assigned within the planning area, following a uniform distribution. Each

[1] For example, if customer $i$ is postponed from day $t$ and delivered on day $s$, then the holding cost w.r.t. the customer is $h_i(s-1-t)$, which the last term of (3.15) represents.

customer initially selects the online shopping frequency and order days according to the frequency choice model presented in Section 3.2.1. Subsequently, on each of the order days, s/he chooses a timing to receive the ordered item based on the delivery timing choice model introduced in Section 3.2.2.

Delivery routes are decided according to the generated demand. At each day $t$, the delivery routes for $d$ days are planned by solving the $d$-MPVRP as described in Section 3.3, given the orders revealed by day $t-1$. Then the delivery for day $t$ is executed, and the plan on and after day $t+1$ is updated day to day by repeatedly solving the $d$-MPVRP. We used the mathematical optimization solver Gurobi Optimizer[2] to optimize day-to-day delivery plan.

We use two indicators to evaluate demand management strategies:

(i) Profit obtained from delivery operations

Profit is calculated by subtracting the delivery cost $z_t(x^*)$ from the revenue $\sum_n r_{tn}$ generated from the delivery fees paid by customer $n$ on day $t$. Here, $z_t(x^*)$ represents the objective value (3.15) with the optimal solution for the delivery plan on day $t$. When routing period $d$ is greater than 1, the objective value includes the delivery costs on day $t+1$ or later, whereas $z_t(x^*)$ only includes the value on day $t$, i.e. the costs incurred in the actual delivery. The delivery cost includes vehicle costs, transportation costs, and holding costs. Therefore, the profit obtained during the evaluation period is $\sum_{t=10}^{16}\{\sum_n r_{tn} - z_t(x^*)\}$.

(ii) Customer satisfaction

Customer satisfaction is based on the expected maximum utility (EMU), calculated with the estimated delivery timing choice model. The EMU is defined as follows:

$$\mathrm{EMU}_n = \frac{1}{\mu}\ln\left\{\exp(\mu V_{n,\text{next-day}}) + \exp(\mu V_{n,\text{normal}}) + \frac{\mu}{\mu_g}\left(\sum_{i\in S}\exp(\mu_g V_{ni})\right)\right\} \tag{3.23}$$

## 4. Results

### 4.1. Model estimation results

Here, we present estimation results for the two choice models presented in Sections 3.2.1 and 3.2.2.

#### 4.1.1. Online shopping frequency model

The estimation results are shown in Table 3.[3] Overall, findings from our model estimation results using Japanese survey data are consistent with those of other studies conducted in different countries. It is found that the online shopping frequency is higher in urban areas where the share of private car use is low, due to the superior convenience of private cars over public transportation for shopping, consistent with the result reported by Dias et al. (2020). The lower frequencies for those over 50 and under 30 are consistent with the model estimation results presented by Reiffer et al. (2023). The table also shows that the frequency is higher for singles and lower for men, consistent with Saphores and Xu (2021) and Pani et al. (2024).

Additionally, the presence of delivery lockers, which eliminates the need for users to be at home to receive goods, encourages a more active use of online shopping. Furthermore, membership programs such as Amazon Prime, which offers free shipping, have also been found to increase online shopping frequency. Another new finding is a higher frequency of online shopping of teleworkers, which rapidly increased and became established during and after the COVID-19 pandemic.

**Table 3**
Estimation results of the online shopping frequency model.

| Explanatory variable | Estimate | t-stat |
|---|---|---|
| Modal share of private cars | −0.633 | −3.41** |
| ≥50 years old | −0.261 | −4.39** |
| <30 years old | −0.624 | −3.96** |
| Single | −0.134 | −1.94* |
| Owns a delivery box | 0.259 | 4.06** |
| Male | −0.227 | −3.93** |
| Free shipping member | 0.503 | 9.04** |
| Telework at least once a week | 0.564 | 8.92** |
| $\kappa_1$ | −0.604 | −7.45** |
| $\kappa_2$ | 1.316 | 50.2** |
| $\kappa_3$ | 3.496 | 26.9** |
| Sample size | | 4721 |
| Initial log-likelihood | | −6626.99 |
| Final log-likelihood | | −5285.62 |
| Adjusted rho-square | | 0.202 |
| AIC | | 10593.23 |
| BIC | | 10664.29 |

**Table 4**
Estimation results of the delivery timing choice model.

| | MNL | | NL | |
|---|---|---|---|---|
| Parameter | Estimate | t-stat | Estimate | t-stat |
| $\alpha_{\text{next-day}}$ | −0.283 | −8.97** | −0.041 | −1.03 |
| $\alpha_{\text{scheduled}}$ | −2.270 | −30.40** | −1.564 | −14.69** |
| $\alpha_{\text{morning}}$ | 0.449 | 16.01** | 0.344 | 13.40** |
| $\alpha_{\text{night}}$ | 0.269 | 8.24** | 0.275 | 10.90** |
| $\alpha_{\text{holiday}}$ | 0.499 | 13.59** | 0.450 | 16.29** |
| $\beta_{\text{day}}$ | −0.405 | −51.40** | −0.319 | −26.95** |
| $\beta_{\text{fee}}$ | −0.009 | −83.15** | −0.009 | −79.76** |
| $\beta_{\text{window}}$ | −0.068 | −2.56* | −0.054 | −2.58** |
| $\mu_g$ | | | 1.305 | 28.86** |
| Sample size | | 19,860 | | 19,860 |
| Initial log-likelihood | | −62270.92 | | −62270.92 |
| Final log-likelihood | | −35630.30 | | −35603.70 |
| Adjusted rho-square | | 0.4278 | | 0.4282 |
| AIC | | 71276.60 | | 71225.39 |
| BIC | | 71339.77 | | 71296.46 |
| Likelihood-ratio test | $-2\{L(\hat{\beta}_{\text{MNL}}) - L(\hat{\beta}_{\text{NL}})\} = 53.20 > 3.84 = \chi^2_{1,0.05}$ | | | |

#### 4.1.2. Delivery timing choice model

Table 4 reports the estimation results of the delivery timing choice model. Both MNL and NL models have the same signs for the coefficients of the delivery attributes, and the likelihood ratio test suggests the statistical preference for the NL model over the MNL model with a 95% confidence level. The scale parameter $\mu_g$ was estimated to be 1.305, which gives $1/\mu_g = 0.766$, confirming an appropriate nest structure. The VODT for EC users, estimated by $\beta_{\text{day}}/\beta_{\text{fee}}$, is 35.4 JPY/day (0.22 USD/day at the rate of 160 JPY to 1 USD). This result describes the customer's willingness to pay for a shorter delivery day, implying that, on average, customers are willing to wait one additional day if the delivery fee is about 35 JPY higher.[4]

Moreover, the estimation result suggests that an increase in the range of a time slot decreases the utility of scheduled delivery. This is because customers do not want to wait long at home to receive their ordered items. The estimate of $\alpha_{\text{holiday}}$ suggests that delivery on holidays is preferred.

Because the NL model has a better model fit than the MNL model, we use the NL model for the delivery timing choice simulation in the following results.

[2] https://www.gurobi.com

[3] We made the most of the available individual attributes in the dataset and reached the final model that is the most interpretable. Some important socio-economic attributes, such as income, were not available.

[4] A more detailed analysis of VODT can be found in Oyama et al. (2024) that estimate mixed logit models, but as this study focuses on a simulation-based evaluation of demand management strategies, we keep the choice model and VODT analysis simple.

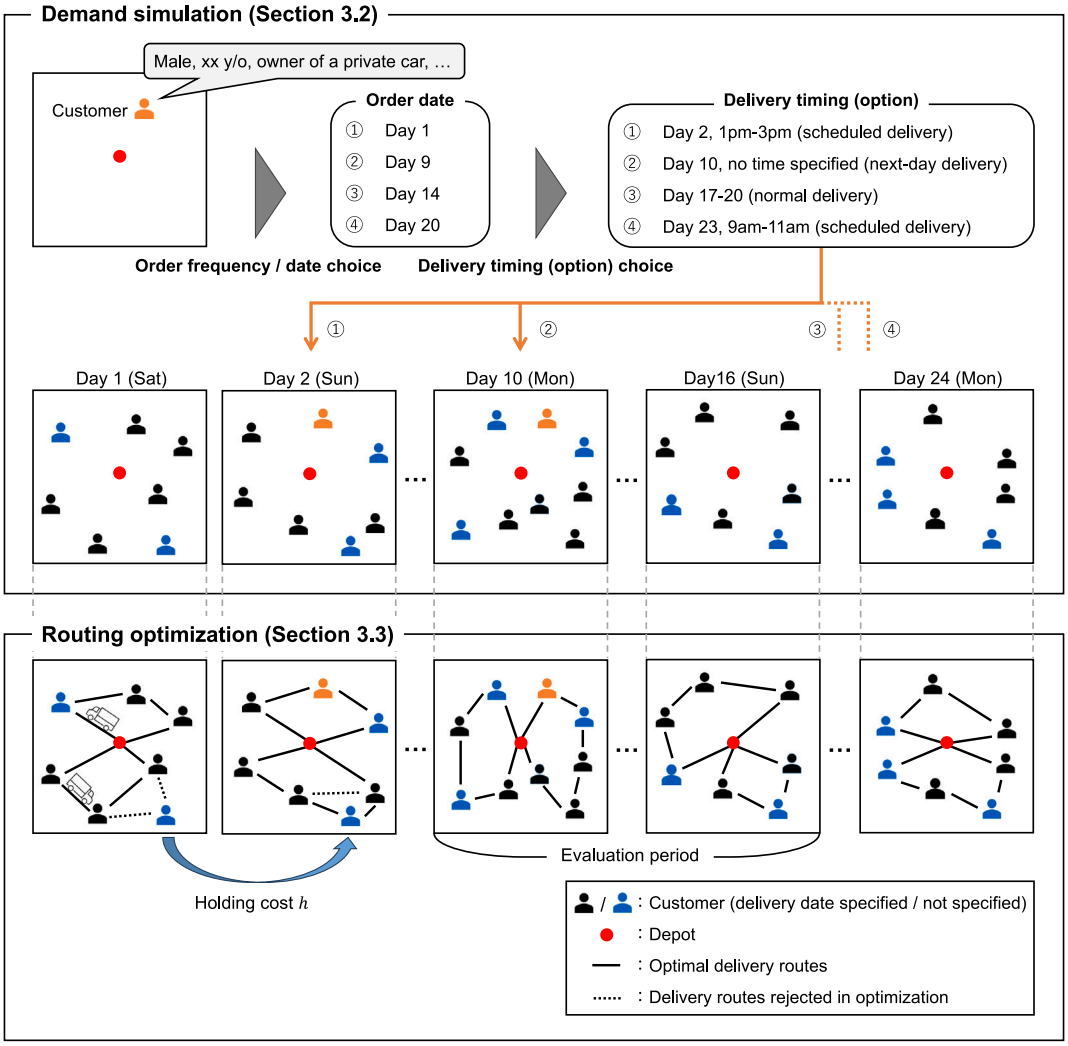



**Fig. 2.** Overview of evaluation framework.

### 4.2. Routing policy comparison

Next, we evaluate the routing policies [$d$-MPVRP] presented in Section 3.3. Specifically, we compare three different routing periods $d$ of 1, 2, and 24 (= $T$) from the viewpoint of delivery costs. When $d = 1$ and $d = 2$, customer orders arrive on a day-to-day basis, and the courier optimizes the routes for $d$ days every day in a rolling horizon manner. In contrast, when $d = 24$, all demand for the entire planning period is known at the beginning, which means that we solve a static MPVRP (Archetti et al., 2015; Larrain et al., 2019). We call this routing policy a "global" policy, serving as a benchmark to evaluate the other policies.

For evaluation, we set the initial cost of a vehicle $v$ to 600, the maximum number of vehicles $m$ to 5, the capacity of a vehicle $Q$ to 200, and all vehicles are homogeneous. The demand quantity $q_i$ is randomly drawn from $[10, 30]$, and the holding cost $h_i$ is equal to $q_i$. In the simulation, customers are generated in a square 100 on one side, and a single depot is located in its center. We generate the same number of customers (20 customers) each day (i.e., we do not use a frequency model) in all policies to eliminate the effect of demand, and the generated customers are common for all policies. Regarding the delivery attributes, we set the time slot size to 2 h, and all delivery options to 400 JPY.

Fig. 3 compares the weekly delivery costs of different routing policies. Fig. 3(a) shows an overall comparison of the three policies for 100 instances, and the delivery cost tends to be Global $< d = 2 < d = 1$. As expected, the delivery cost of the global policy is the minimum because it is an ideal plan for which the entire demand in the planning period is all known.

Fig. 3(b)–(d) are the pairwise comparisons of policies plotting the delivery costs for each instance, with lines drawn where the delivery costs for the two policies are equal. Fig. 3(b) shows that $d = 2$ is superior to $d = 1$ (smaller delivery cost) in almost all instances. For the two policies ($d = 1$ and $d = 2$) where orders dynamically arrive, the decision of whether to deliver a package for regular delivery on that day or postpone it to the next day affects the delivery cost. The policy with $d = 2$ allows us to consider the distribution of customers on the next day, making it easier to plan a more efficient routing than in the $d = 1$ policy. However, for some instances, the delivery costs of the $d = 2$ policy are larger than the $d = 1$ policy, due to uncertainty in order arrival.

From Fig. 3(c) and (d), we can compare how inferior $d = 1$ and $d = 2$ are to Global. The policy with $d = 2$ is clearly closer to Global than that with $d = 1$. In many instances, the delivery costs of $d = 2$ and Global are almost equivalent. This result indicates that by considering only the next day, the delivery route can be made much more efficient and almost as efficient as the global routing policy that considers all demand in the entire planning period.

Thus, in the following, we use the $d = 2$ routing policy for evaluating the demand management strategies.

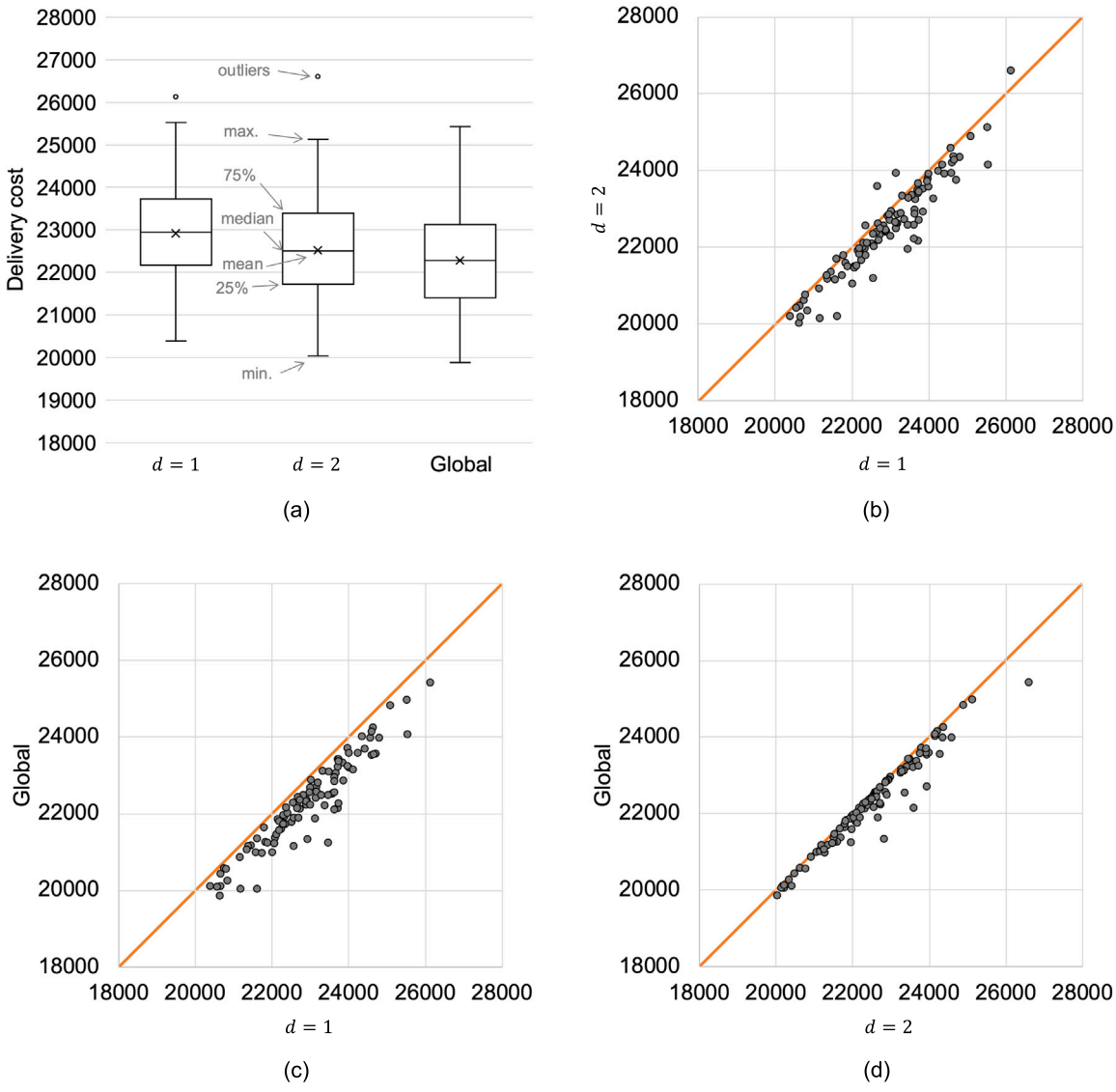



**Fig. 3.** Comparison of delivery costs for each routing policy.

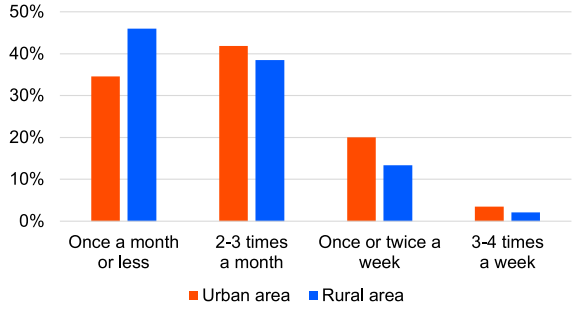



**Fig. 4.** Regional comparison of online shopping frequency estimated from the model.

### 4.3. Evaluation of demand management strategy

Finally, we evaluate the following strategies that can change the delivery timing choice behavior of customers. In the benchmark, we set the delivery fee for all delivery options to 400 JPY.

1: Additional charge of 100 JPY for next-day delivery
In the SP survey, approximately 45% of respondents chose next-day delivery. However, as argued by Rai et al. (2019) and Oyama et al. (2024), many users choose next-day delivery even if it is not necessary. In addition, the nature of next-day delivery means that orders are not completed until the day before the delivery date, which may make day-to-day delivery planning more difficult. Therefore, we charged an additional charge of 100 JPY for next-day delivery to disperse demand.

2: Additional charge of 100 JPY for delivery on holidays
As Table 2 shows, the choice rate for holiday delivery is much higher than that for weekdays. Therefore, we raise the delivery fee on holidays by 100 JPY in order to induce demand on weekdays.

3: Additional charge of 100 JPY for delivery in the morning slot
As Table 2 shows, the choice rate in the morning is nearly half, therefore we charge 100 JPY for the earliest time slot in each day.

4: Discount of 100 JPY on normal delivery for customers who shop online frequently
In this strategy, we focus on the order frequency of each customer. Customers who shop online frequently force the courier to make multiple round trips between the depot and their home.

**Table 5**
Comparison of numerical results under each strategy. Cost/Customer is the delivery cost per customer and CS is customer satisfaction (average per customer). Gap (%) represents the profit gap compared to the benchmark for each slot length, with the asterisk indicating statistical significance at the 95% level based on a two-sample t-test.

| Urban area | | | | | | | |
|---|---|---|---|---|---|---|---|
| Strategy | Slot length | Revenue | Delivery cost | Profit ± StdDev | Gap (%) | Cost/Customer | CS |
| Benchmark | 2 h | 64,408 | 22,439 | 41969 ± 1590 | – | 138.9 | −3.39 |
| | 4 h | 64,431 | 21,811 | 42620 ± 1645 | – | 134.9 | −3.40 |
| Next-day +100 JPY | 2 h | 68,426 | 23,892 | 44534 ± 1776 | 6.1* | 147.7 | −3.88 |
| | 4 h | 68,670 | 23,082 | 45588 ± 1677 | 7.0* | 143.3 | −3.91 |
| Weekend +100 JPY | 2 h | 68,188 | 22,052 | 46136 ± 1739 | 9.9* | 136.5 | −3.60 |
| | 4 h | 68,175 | 21,443 | 46732 ± 1654 | 9.6* | 132.6 | −3.62 |
| Morning +100 JPY | 2 h | 65,372 | 20,717 | 44655 ± 1696 | 6.4* | 127.9 | −3.43 |
| | 4 h | 65,346 | 20,137 | 45209 ± 1618 | 6.1* | 125.0 | −3.44 |
| Normal −100 JPY | | | | | | | |
| $\alpha = 5$ | 2 h | 58,321 | 20,876 | 37445 ± 1490 | −10.8* | 129.4 | −3.13 |
| | 4 h | 58,350 | 20,278 | 38072 ± 1446 | −10.7* | 125.5 | −3.15 |
| $\alpha = 10$ | 2 h | 58,511 | 20,977 | 37534 ± 1467 | −10.6* | 130.3 | −3.15 |
| | 4 h | 58,535 | 20,473 | 38062 ± 1484 | −10.7* | 126.9 | −3.16 |
| $\alpha = 15$ | 2 h | 59,403 | 21,286 | 38117 ± 1505 | −9.2* | 131.4 | −3.17 |
| | 4 h | 59,397 | 20,723 | 38674 ± 1519 | −9.3* | 128.4 | −3.18 |
| $\alpha = 20$ | 2 h | 60,866 | 21,582 | 39284 ± 1398 | −6.4* | 134.0 | −3.24 |
| | 4 h | 60,807 | 21,029 | 39778 ± 1491 | −6.7* | 130.3 | −3.26 |
| $\alpha = 25$ | 2 h | 62,185 | 21,863 | 40323 ± 1617 | −3.9* | 135.6 | −3.30 |
| | 4 h | 62,107 | 21,174 | 40933 ± 1502 | −4.0* | 131.3 | −3.33 |
| $\alpha = 30$ | 2 h | 63,247 | 22,068 | 41179 ± 1593 | −1.9* | 137.0 | −3.34 |
| | 4 h | 63,224 | 21,503 | 41721 ± 1677 | −2.1* | 133.1 | −3.37 |
| $\alpha = 35$ | 2 h | 63,739 | 22,206 | 41533 ± 1561 | −1.0* | 137.6 | −3.35 |
| | 4 h | 63,732 | 21,634 | 42098 ± 1549 | −1.2* | 134.1 | −3.38 |

| Rural area | | | | | | | |
|---|---|---|---|---|---|---|---|
| Strategy | Slot length | Revenue | Delivery cost | Profit ± StdDev | Gap (%) | Cost/Customer | CS |
| Benchmark | 2 h | 50,390 | 30,723 | 19667 ± 714 | – | 241.7 | −3.39 |
| | 4 h | 50,382 | 28,650 | 21732 ± 823 | – | 225.4 | −3.40 |
| Next-day +100 JPY | 2 h | 53,562 | 33,190 | 20372 ± 778 | 3.6* | 262.6 | −3.88 |
| | 4 h | 53,746 | 30,523 | 23223 ± 852 | 6.9* | 239.6 | −3.91 |
| Weekend +100 JPY | 2 h | 53,344 | 30,827 | 22517 ± 812 | 14.5* | 244.3 | −3.60 |
| | 4 h | 53,398 | 28,675 | 24723 ± 937 | 13.8* | 227.2 | −3.62 |
| Morning +100 JPY | 2 h | 51,153 | 30,181 | 20972 ± 788 | 6.6* | 236.2 | −3.43 |
| | 4 h | 51,084 | 28,145 | 22939 ± 910 | 5.6* | 221.0 | −3.44 |
| Normal −100 JPY | | | | | | | |
| $\alpha = 5$ | 2 h | 46,136 | 26,732 | 19404 ± 737 | −1.3* | 209.0 | −3.13 |
| | 4 h | 46,101 | 25,402 | 20699 ± 840 | −4.8* | 199.3 | −3.15 |
| $\alpha = 10$ | 2 h | 46,778 | 26,906 | 19872 ± 792 | 1.0* | 211.9 | −3.18 |
| | 4 h | 46,761 | 25,564 | 21197 ± 754 | −2.5* | 199.7 | −3.19 |
| $\alpha = 15$ | 2 h | 48,710 | 27,873 | 20837 ± 831 | 5.9* | 220.0 | −3.28 |
| | 4 h | 48,692 | 26,413 | 22279 ± 784 | 2.5* | 206.5 | −3.30 |
| $\alpha = 20$ | 2 h | 49,496 | 29,353 | 20143 ± 763 | 2.4* | 232.3 | −3.33 |
| | 4 h | 49,429 | 27,565 | 21864 ± 870 | 0.6* | 216.3 | −3.36 |
| $\alpha = 25$ | 2 h | 50,041 | 30,262 | 19779 ± 729 | 0.6* | 238.1 | −3.37 |
| | 4 h | 49,991 | 28,352 | 21639 ± 805 | −0.4* | 223.6 | −3.39 |

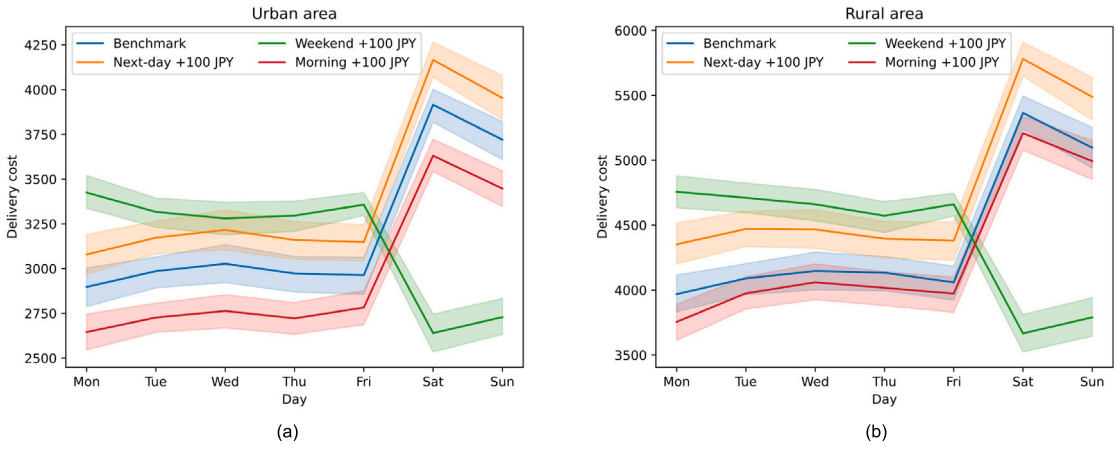



**Fig. 5.** Day-to-day delivery costs under each strategy (slot length: 2 h).

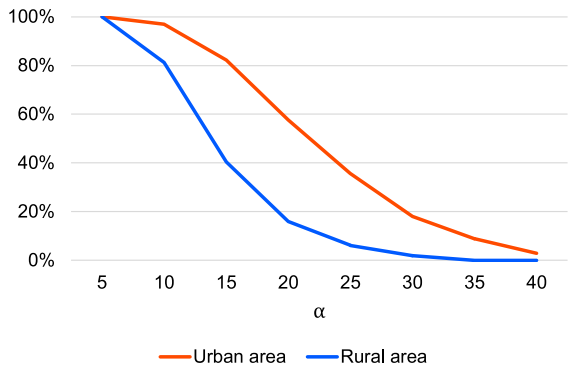



**Fig. 6.** The proportion of customers whose probability of ordering at least once a week exceeds $\alpha$ (%).

Therefore, with this strategy, we incentivize high-frequency customers to choose normal delivery so that the courier can consolidate multiple delivery orders of the same customer in a short period into a single delivery. The model estimated in 3.2.1 allows us to calculate the probability of choosing the online shopping frequency for each customer. We discount normal delivery by 100 JPY for each customer whose probability of shopping online more than once a week exceeds $\alpha$%. We also vary the threshold value $\alpha$ and analyze the changes in profit and customer satisfaction.

5: Change the range of time slots from 2 h to 4 h

Most scheduled delivery services currently offered in Japan have a time slot of 2 h. In addition to this, we evaluate the cases with a time slot length of 4 h. A longer time slot relaxes the time window constraints for delivery to be satisfied in finding the optimal route, thus improving delivery efficiency. However, a longer slot length increases the uncertainty of the delivery time, which is expected to decrease customer satisfaction. We combine this strategy with the above four strategies.

Since last-mile small-sized facilities have a spatial distribution that follows population and business establishment density (Sakai et al., 2024), the effectiveness of demand management strategies may differ depending on differences in customer density and the size of a delivery service area for each depot. Therefore, we analyze the impact of the strategies in different areas with different geographical characteristics, i.e., urban and rural areas. The two areas differ in terms of the area size as well as the frequency choice probabilities due to the mode share. One side of the square is 100 in urban areas and 200 in rural areas. That is, the area of the rural area is four times larger than the urban area to consider the low density of the rural area. This ratio of area sizes (1:4) is determined to reflect the realism of the difference delivery service areas of facilities in urban and rural areas in the Tokyo metropolitan area (Sakai et al., 2024) and to effectively analyze the impact of such a difference on routing problems.

For simulation, we generate 20 customers each day as in the routing policy comparison in Section 4.2. We randomly assign individual attributes to each customer according to the following statistics. First, we set the modal share of private cars in urban areas at 8% and the modal share of private cars in rural areas at 70% based on the Tokyo Person Trip Survey in 2018[5] and the report of the Japanese Ministry of Land, Infrastructure, Transport and Tourism in 2021,[6] respectively. The distributions of the other attributes are based on data published by the Statistics Bureau of Japan.[7] For example, Fig. 4 shows the difference in the predicted frequency choice probabilities between urban and rural areas. The frequency is higher in urban areas where the modal share of private cars is lower.

Tables 5 compare the strategies both in urban and rural areas. Regarding the benchmarks, the urban area has higher revenues than the rural area because of the higher online shopping frequency. Since demand is more sparsely distributed in rural areas, delivery costs tend to be higher, which is consistent with the finding of Pahwa and Jaller (2022). Thus, profit differs significantly between the two areas. The average customer satisfaction is the same in urban and rural areas because all customers' delivery fees are set at the same price. Focusing on the effect of increasing the time slot length to four hours, the reduction in delivery costs is greater in rural areas (6.7%) than in urban areas (2.8%). This is also due to the sparseness of customers in the rural areas. Interestingly, increasing the time slot length has little effect on customer satisfaction compared to the additional charges.

Next, we report on the effects of the three surcharge strategies 1–3. Of the three, only Morning +100 JPY contributed to the reduction of delivery costs in both regions. Weekend +100 JPY reduced delivery costs only in urban areas. Surcharges for time slots with concentrated demand, such as Morning+100 JPY would be more effective than surcharges for "days" such as Next-day or Weekend +100 JPY. Also, the amount of decrease in customer satisfaction is smaller for the Morning +100 JPY compared to the other surcharge strategies. Fig. 5 depicts the daily delivery costs under different surcharge strategies as well as the benchmark. This figure clearly shows that the Weekend +100 JPY shifted demand to weekdays and successfully reduced the cost on weekends.

We then look at the effect of Normal −100 JPY (Strategy 4), which is the strategy focusing on online shopping frequency. Fig. 6 shows the percentage of customers offered discounts for different threshold values of $\alpha$. The solid line in the graph shows the interval where we performed numerical experiments. If $\alpha$ is set at an extremely low value (5%), then all customers will receive a 100 JPY discount on normal delivery in both areas. Also, setting $\alpha = 35$ (%) in the urban area and $\alpha = 25$ (%) or more in the rural area will result in few customers receiving discounts and thus the same result as the benchmark. Fig. 7 reports the changes in revenue, delivery costs, profit, and customer satisfaction with the value of $\alpha$. Most importantly, profits in the rural areas are higher than the benchmark when $\alpha = 10$–$20$(%) even though we offer discounts to customers. This is because the reduction in delivery costs is greater than the reduction in revenue. The smaller the value of $\alpha$, the more customers choose normal delivery. Normal delivery not only offers flexibility for the courier in delivery timing, but also the possibility of consolidating multiple orders from the same customer into a single delivery, leading to a great reduction in delivery cost. The result also shows a greater impact of this strategy in the rural area than in the urban area. In fact, delivery in rural areas requires more cost per customer due to low population density, and therefore, the benefit of flexibility/consolidation of delivery is high. The average customer satisfaction increases as the number of customers offered discounts increases, and thus the change in customer satisfaction follows a curve similar to that of the percentage of customers offered discounts in Fig. 6.

## 5. Discussion

This study presented a quasi-dynamic MPVRP in which customers' orders appear day-to-day and the route planning is re-optimized accordingly. The original MPVRP by Archetti et al. (2015) considers all demand during the entire planning period to be known at the beginning and thus can optimize the route planning. The routing policy

[5] https://www.tokyo-pt.jp/person/01 (Japanese).

[6] https://www.mlit.go.jp/report/press/content/001485333.pdf (Japanese).

[7] https://www.stat.go.jp/english/index.html

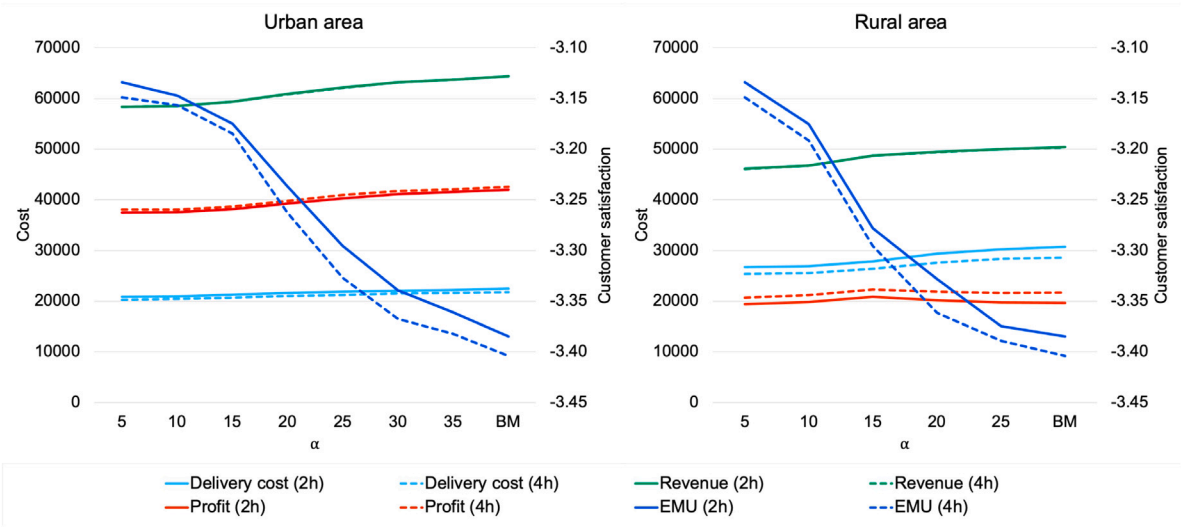



**Fig. 7.** Change in revenue, delivery cost, and profit with value of $\alpha$ (%).

comparison in Section 4.2 showed that the routing result of our $d = 2$ policy performs in most cases as well as the original MPVRP, indicating that day-to-day delivery planning can be made more efficient simply by considering the next day's demand in the routing. Considering that the future demand for day-to-day home delivery is highly uncertain, the fact that quasi-dynamic routing with a small routing period $d$ performs as well as the global routing policy may provide a simple and efficient routing strategy for courier companies.

As for the evaluation of demand management strategies, although charging an additional fee on certain days (the next day or holidays) effectively shifts demand to other days, it does not always reduce delivery costs. Particularly, the strategy to charge an additional 100 JPY for next-day delivery increases delivery costs considerably because it shifts demand to scheduled delivery, which costs more. However, it is important to note that this study only considers the costs of last-mile delivery, and the evaluation of a strategy to reduce demand for next-day delivery should take into account the upstream processes of the supply chain. For instance, reducing the share of next-day delivery orders through pricing strategies may alleviate operational pressures in upstream processes by relaxing the lead-time constraint. Although this study does not explicitly model these upstream effects, the delivery timing choice model provides a foundation for future research to evaluate such impacts across the entire supply chain.

While charging additional fees has some effect in urban areas, it is less effective in rural areas. In contrast, the strategy focusing on online shopping frequency is particularly effective in rural areas, achieving an increase in both profits and customer satisfaction. This is possible because the shift of customers' delivery option choice to normal delivery gives a courier the flexibility of delivery timing and the possibility of consolidating multiple orders from the same customer into a single delivery. Furthermore, it was found ineffective to offer discounts to all customers, and targeting only frequent online shoppers with a greater than 15% probability of shopping online at least once a week is particularly effective in terms of profit. These findings suggest that demand management strategies have to be chosen considering the spatial and temporal distribution of demand. Note that delivery bundling for high-frequency customers has already been implemented in real-world e-marketplaces (e.g., *Amazon Day*), and our findings indicate that price-based incentives focusing on order frequency can effectively promote such bundling through the choice of normal delivery. At the same time, since customers themselves can benefit from bundled deliveries, our future research will explore alternative non-price incentives (e.g., green labeling or slot offering) for practical applications.

We also compared different time slot lengths of 2 h and 4 h and found that a wider time slot reduces routing costs, confirming the consistency with previous results (Agatz et al., 2011; Köhler et al., 2020; Pahwa & Jaller, 2022). In addition, we find that this effect is particularly significant in rural areas where demand is sparse and that a wider time slot length has a smaller impact on customer satisfaction compared to additional charges. The reason for this can be seen from the estimation results of the delivery timing choice model shown in Table 4, where the sensitivity to an increase in the time slot length is smaller than that to an increase in the delivery fee, which is consistent with Oyama et al. (2024).

Here are some remarks on the limitations of the analysis. First, in the experiments, we generate 20 customers every day. Increasing the number of customers leads to more loads on specific days or time slots, which may violate the vehicle capacity constraint. In such situations, strategies to promote normal delivery might become more effective. The size of demand may also impact the holding strategy of day-to-day delivery. Although the MPVRP model optimizes route planning for multiple days by postponing some delivery requests, the current setting assumes unlimited storage capacity of the depot. Its limited capacity with large delivery demand may lead to a higher holding cost and thus less efficient routing. A detailed investigation of the impacts of these parameters will be necessary for practical applications. Second, while we use an ordered logit model and an NL model, substantial evidence in the literature indicates significant heterogeneity in e-commerce delivery preferences (e.g., Oyama et al., 2024). The use of advanced models (e.g., mixed logit or latent class models) allows us to flexibly capture unobserved heterogeneity and evaluate more sophisticated strategies, such as the personalization of delivery menus. The integration of advanced models into our evaluation framework requires efficient computational methods, but investigating the effect of heterogeneous preferences on the efficiency of the proposed strategies is an important next step. Third, the present delivery timing choice model does not consider an opt-out option for customers to cancel the delivery order. Even though we use customer satisfaction as an indicator, we cannot discuss the impact of its decline on revenue. Therefore, future work will integrate the opt-out option into our evaluation framework, e.g., by modeling cancelation behavior or shopping mode/market switching. The opt-out probability is also necessary information for optimal pricing (e.g., Abdollahi et al., 2023; Yang & Strauss, 2017).

## 6. Conclusions

This study presented a framework for evaluating choice-based demand management strategies for day-to-day home delivery planning.

The framework aims to balance customer satisfaction and delivery efficiency for sustainable last-mile delivery. We estimated two behavior models of EC customers, namely the online shopping frequency model and the delivery timing choice model, based on which we simulated day-to-day home delivery demand. We also presented a quasi-dynamic MPVRP ($d$-MPVRP) for a novel route planning policy. Based on this framework, we evaluated different demand management strategies, including time-window expansion, surcharges, and frequency-focused discounting. We also discussed effective strategies both in urban and rural areas, focusing on differences in demand density and order frequency. These findings can contribute to designing demand management strategies that improve operational efficiency in last-mile delivery while maintaining customer satisfaction. As such, this study provides valuable insights for effective delivery demand management in the real-world e-commerce market.

## CRediT authorship contribution statement

**Ryota Okazaki:** Writing – original draft, Methodology, Data curation, Visualization, Investigation, Software, Formal analysis. **Yuki Oyama:** Writing – review & editing, Resources, Formal analysis, Writing – original draft, Methodology, Data curation, Supervision, Investigation, Conceptualization. **Naoto Imura:** Resources. **Katsuhiro Nishinari:** Resources.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix. Impact of increase in delivery box ownership

Delivery boxes have been shown effective in reducing delivery costs, and their number is growing worldwide (e.g., Ranjbari et al., 2023; Schaefer & Figliozzi, 2021). Our analysis in Table 3 shows that

**Table A.6**
Strategy evaluation results with the delivery box penetration rate of 75%.

Urban area

| Strategy | Slot length | Revenue | Delivery cost | Profit ± StdDev | Gap (%) | Cost/Customer | CS |
|---|---|---|---|---|---|---|---|
| Benchmark | 2 h | 73,425 | 22,865 | 50560 ± 1593 | 0.0%* | 135.5 | −3.39 |
| | 4 h | 73,474 | 22,509 | 50965 ± 1674 | 0.0%* | 133.3 | −3.40 |
| Next-day +100 JPY | 2 h | 75,953 | 24,585 | 51368 ± 1595 | 1.6%* | 145.5 | −3.88 |
| | 4 h | 75,884 | 23,589 | 52295 ± 1764 | 2.6%* | 140.2 | −3.91 |
| Weekend +100 JPY | 2 h | 75,643 | 22,559 | 53084 ± 1784 | 5.0%* | 133.7 | −3.60 |
| | 4 h | 75,583 | 21,979 | 53604 ± 1806 | 5.2%* | 130.1 | −3.62 |
| Morning +100 JPY | 2 h | 74,409 | 21,007 | 53402 ± 1692 | 5.6%* | 124.2 | −3.43 |
| | 4 h | 74,348 | 20,842 | 53506 ± 1859 | 5.0%* | 123.9 | −3.45 |
| Normal −100 JPY | | | | | | | |
| $\alpha = 5$ | 2 h | 67,069 | 21,377 | 45692 ± 1411 | −9.6%* | 126.9 | −3.10 |
| | 4 h | 66,902 | 20,907 | 45995 ± 1507 | −9.8%* | 123.9 | −3.12 |
| $\alpha = 10$ | 2 h | 64,362 | 21,542 | 42820 ± 1461 | −15.3%* | 128.1 | −3.12 |
| | 4 h | 64,389 | 20,922 | 43466 ± 1400 | −14.7%* | 124.2 | −3.14 |
| $\alpha = 15$ | 2 h | 65,344 | 21,967 | 43377 ± 1523 | −14.2%* | 129.9 | −3.14 |
| | 4 h | 65,525 | 20,991 | 44533 ± 1535 | −12.6%* | 124.6 | −3.16 |
| $\alpha = 20$ | 2 h | 67,562 | 22,186 | 45376 ± 1540 | −10.3%* | 131.9 | −3.19 |
| | 4 h | 67,580 | 21,491 | 46089 ± 1535 | −9.6%* | 127.5 | −3.21 |
| $\alpha = 25$ | 2 h | 70,269 | 22,409 | 47861 ± 1684 | −5.3%* | 133.1 | −3.26 |
| | 4 h | 70,176 | 21,915 | 48261 ± 1520 | −5.3%* | 130.1 | −3.28 |
| $\alpha = 30$ | 2 h | 71,307 | 22,743 | 48564 ± 1640 | −3.9%* | 133.4 | −3.29 |
| | 4 h | 71,314 | 22,347 | 48967 ± 1657 | −3.9%* | 131.3 | −3.31 |
| $\alpha = 35$ | 2 h | 72,300 | 23,061 | 49239 ± 1555 | −2.6%* | 134.2 | −3.31 |
| | 4 h | 72,292 | 22,715 | 49577 ± 1664 | −2.7%* | 130.1 | −3.34 |

Rural area

| Strategy | Slot length | Revenue | Delivery cost | Profit ± StdDev | Gap (%) | Cost/Customer | CS |
|---|---|---|---|---|---|---|---|
| Benchmark | 2 h | 51,902 | 31,706 | 20196 ± 834 | 0.0%* | 238.9 | −3.39 |
| | 4 h | 51,933 | 29,510 | 22423 ± 850 | 0.0%* | 222.3 | −3.40 |
| Next-day +100 JPY | 2 h | 55,704 | 33,953 | 21751 ± 870 | 7.7%* | 257.2 | −3.88 |
| | 4 h | 55,564 | 31,591 | 23974 ± 947 | 6.9%* | 237.5 | −3.91 |
| Weekend +100 JPY | 2 h | 55,070 | 31,598 | 23472 ± 965 | 16.2%* | 239.8 | −3.60 |
| | 4 h | 55,083 | 29,621 | 25462 ± 963 | 13.6%* | 224.8 | −3.62 |
| Morning +100 JPY | 2 h | 53,045 | 31,026 | 22019 ± 909 | 9.0%* | 232.5 | −3.43 |
| | 4 h | 52,935 | 29,017 | 23918 ± 918 | 6.7%* | 218.2 | −3.44 |
| Normal −100 JPY | | | | | | | |
| $\alpha = 5$ | 2 h | 47,059 | 27,133 | 19926 ± 803 | −1.3%* | 190.7 | −3.10 |
| | 4 h | 47,173 | 26,113 | 21060 ± 842 | −6.1%* | 196.2 | −3.12 |
| $\alpha = 10$ | 2 h | 48,520 | 27,864 | 20657 ± 822 | 2.3%* | 210.3 | −3.13 |
| | 4 h | 48,710 | 26,378 | 22332 ± 815 | −0.4%* | 195.9 | −3.15 |
| $\alpha = 15$ | 2 h | 49,633 | 28,681 | 20951 ± 869 | 3.7%* | 216.8 | −3.23 |
| | 4 h | 49,451 | 26,809 | 22642 ± 879 | 1.0%* | 200.7 | −3.25 |
| $\alpha = 20$ | 2 h | 50,981 | 29,322 | 21659 ± 886 | 7.2%* | 225.8 | −3.28 |
| | 4 h | 51,099 | 27,979 | 23120 ± 869 | 3.1%* | 210.2 | −3.30 |
| $\alpha = 25$ | 2 h | 51,544 | 30,988 | 20556 ± 810 | 1.8%* | 233.5 | −3.34 |
| | 4 h | 51,619 | 29,174 | 22445 ± 875 | 0.1%* | 220.3 | −3.36 |

customers who have delivery boxes order more frequently. Here, we show how the ever-increasing number of delivery boxes affects last-mile delivery. We herein assume a 75% of penetration rate of delivery boxes, which triples that assumed in the experiments of Section 4.3. An increase in the adoption of delivery boxes increases the total number of orders. Specifically, the average numbers of orders in urban and rural areas are 25.5 and 19.8 per day, respectively, which increased from 22.1 and 19.0 in the case with a penetration rate of 25% of the delivery box. Thus, as shown in Table A.6, revenues increased compared to the case with a lower penetration rate of delivery boxes. Changes in other indicators, such as delivery cost and customer satisfaction, are trivial, especially in the urban area. These trends are similar with all strategies. It should be noted that, in reality, the delivery cost is expected to be lower than this result because the delivery box reduces the risk of delivery failure and redelivery. The presence of a delivery box may affect the delivery timing choice. The investigation of these impacts is an important future work of the study.

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